**Team Caffeinated Nerds** · Challenge: **Frontier · PS-04** · Fast and Secure Online Transaction Authentication

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# Problem, Stakeholder & SDGs

## *> Fraud has moved past OTPs: victims now approve the payment themselves*

### THE PROBLEM: ASHA, 72, PAYS A SCAMMER WITH HER OWN PIN

**Fake "CBI" officer calls** → **Opens UPI on her own phone** → **Types her own PIN** → **Approved: ₹85,000 gone**

A PIN checks **who** you are. Nobody checks **what's happening**.

### WHO WE SERVE

- **Banks, UPI apps & gateways**
- **Seniors & first-time payers**
- **Decide inside UPI's 15-s window**

### EVIDENCE

**₹22,495 cr**
lost to cyber fraud in India in 2025 (MHA)

**~85%**
of losses: scams where the victim pays (MHA)

**98.5%**
of fraud value is in payments above ₹10k (RBI)

**PRIMARY · SDG 16**
16.4: cut illicit financial flows & organised crime

**SUPPORTING · SDG 8**
8.10: access to financial services for all

**SUPPORTING · SDG 9**
9.1: reliable, resilient (digital) infrastructure

*Sources: MHA/NCRP data via Parliament replies (2024, 2025); RBI Discussion Paper on safeguards in digital payments, 9 Apr 2026. Links: repo › docs/02-research/problem-evidence.md*

# Existing Approaches & Gap

## *> Today's defences check the credential or the receiver, never the moment*

| | Static 2FA<br>*UPI PIN · OTP · 3-D Secure* | Risk engines<br>*Stripe Radar · Sift · BioCatch* | India's tools<br>*NPCI FRM · DoT FRI · MuleHunter* | Pramaan<br>*this project* |
|---|---|---|---|---|
| Checks who holds the credential | ✓ | ✓ | ✗ | ✓ |
| Scores the receiver's reputation | ✗ | ✓ | ✓ | ✓ |
| Reads the live moment: call, device, typing | ✗ | — | ✗ | ✓ |
| Explains the decision to the user | ✗ | ✗ | ✗ | ✓ |
| Picks the right challenge for the risk | ✗ | ✗ | ✗ | ✓ |
| Affordable for small banks & apps | ✓ | ✗ | ✓ | ✓ |

✓ yes — partly ✗ no

**The gap:** nothing lightweight reads the moment and answers with an explained, right-sized challenge.

*Sources: RBI (25 Sep 2025, 9 Apr 2026); Stripe Radar, Sift, BioCatch docs; NPCI FRM; DoT FRI; MuleHunter.AI. Links: repo › docs/02-research/prior-art.md*

BIOS

# Proposed Solution

***> Pramaan: a PIN checks who you are. We check what's happening, before every payment.***

## WHAT WE BUILD: A RISK CHECK BEFORE THE PIN

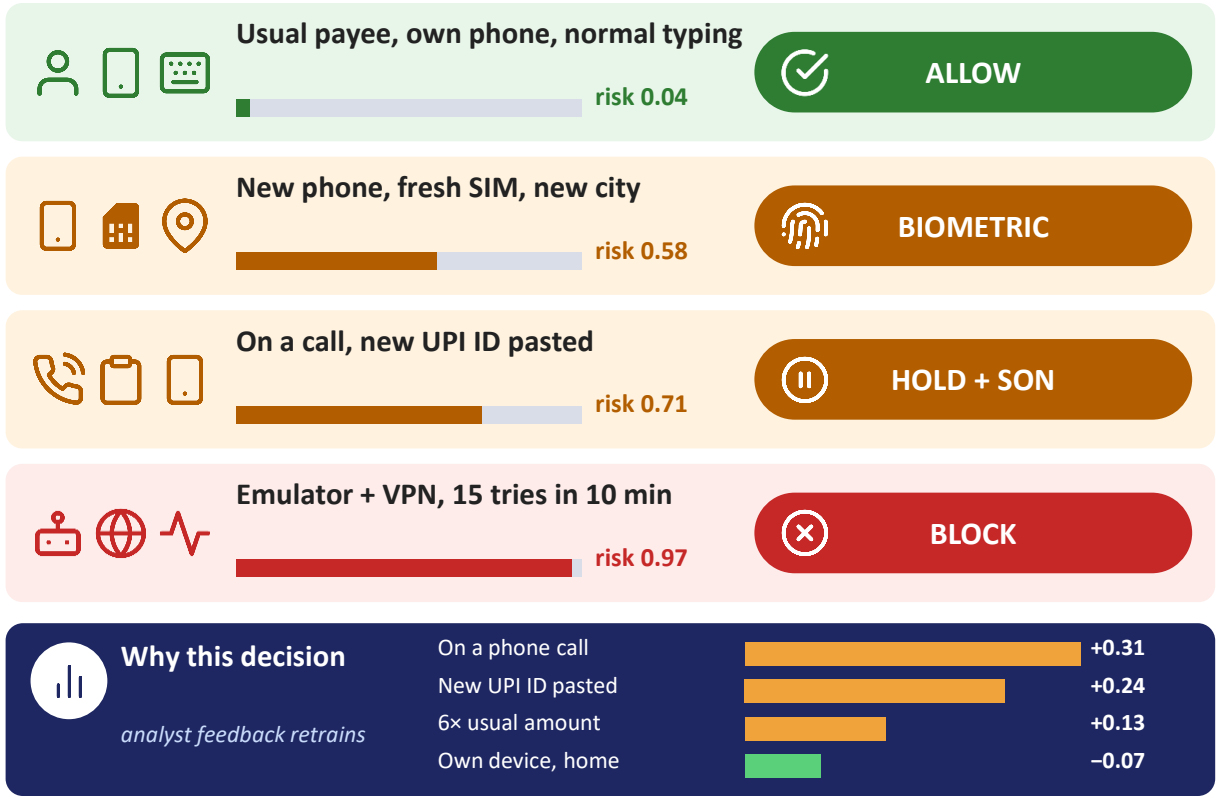
**Bank / UPI app calls us** → **Scores 20+ live signals** → **Decision in ~40 ms**

## WHY IT'S DIFFERENT

- Judged against each user's own habits
- Amount-aware: a ₹40 tea is never stopped
- A plain-language reason with every decision

An OTP can't stop a coached victim. A pause and a family member who can cancel can, **used only when the context is risky.**

## THE SAME ₹85,000 PAYMENT · FOUR SITUATIONS



# Technical Approach

## *> One API call: context in, an explained decision out, in milliseconds*

**1 · INPUT**

**Payment app + SDK**

amount · device · typing · paste · on-call*

**2 · API**

**/v1/authenticate**

FastAPI · over 150 ms → falls back to 2FA

**3 · FEATURES**

**20+ features**

vs the user's own baseline: "8× usual amount"

**4 · RISK ENGINE**

- **a** Rules: hard policy + fast-path allow
- **b** LightGBM → fraud probability p
- **c** TreeSHAP → each signal's weight
- **d** Risk type: impostor · scam · bot
- **e** Cost policy → decision + challenge

**5 · OUTPUT**

- ALLOW
- CHALLENGE
- BLOCK

+ score · reasons · latency

**Built in the hack:** SDK + demo app · API · risk engine · passkeys · n8n + Telegram loop · dashboard **Mocked:** bank/NPCI rails, *call state

### STACK · ₹0 RUNNING COST

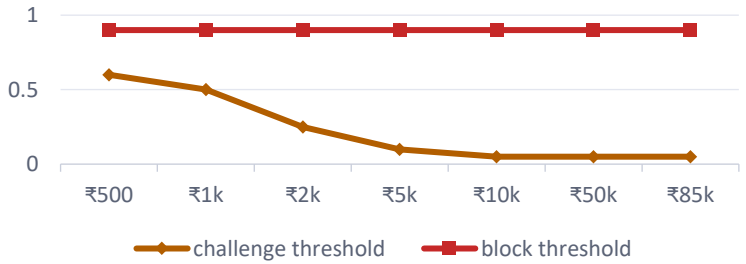
- Python · FastAPI
- LightGBM + SHAP
- React dashboard
- Supabase
- n8n + Telegram
- WebAuthn passkeys

### SYNTHETIC DATA (NO PUBLIC SET HAS THESE SIGNALS)

| 5k | 30 | 4 |
|---|---|---|
| users | days | attack types |

*+ hard negatives (travel, new phone) · checked on IEEE-CIS & PaySim*

### Decision rule: stricter as the amount grows



# Build Plan, Risks & Impact

*> A deployed, working prototype in the hack, honest about its limits*

## 1 · WHAT WE BUILD

- Live API with fail-safe to 2FA
- Rules + LightGBM + TreeSHAP
- Passkey fingerprint check
- Telegram trusted-contact loop
- Dashboard with “why” panel

**24-HOUR PLAN**

| Data | Engine | App + UI | Deploy |
|---|---|---|---|
| H0–6 | H6–12 | H12–18 | H18–24 |

**Cost:** ₹0, free tiers + open source

## 2 · RISKS & HOW WE HANDLE THEM

**Synthetic data**

→ hard negatives, per-type metrics, public-set checks

**Call state needs an SDK**

→ missing = “unknown”, never “safe”

**False positives**

→ block only at p ≥ 0.9; ≤ 5% challenged

**Privacy (DPDP)**

→ derived features only, hashed IDs

## 3 · EXPECTED IMPACT

**Honest payments stay instant.**
**Risky ones get stopped.**

**≥ 80%**
scams caught, ≤ 5% of honest payments challenged

**≥ 90%**
honest ₹10k+ transfers with no delay

**< 50 ms**
p95 decision time (UPI allows 15 s)

*Fast. Secure. Explained.*